

Study Sheet

Reading 2.1–2.2 · Chapter 2: Population and Health · pp. 1–31

Formulas

Natural increase rate

$(\text{CBR} - \text{CDR}) \div 10 = \%$ per year

CBR 35, CDR 10 → 25 per 1,000 → 2.5%

Total population change

natural increase + immigration – emigration

natural increase alone leaves out migration

Doubling time

$70 \div \text{growth } \%$

the math behind the book's \$100 at 10% → ~7 years. 2.5% → about **28 years**; 2% → 35; 1.17% (India) → about 60

Old-age dependency ratio

people 65+ ÷ people 15–64 × 100

12 million ÷ 40 million → **30**

Total dependency ratio

(ages 0–14 + ages 65+) ÷ ages 15–64 × 100

Replacement level

TFR of **2.1** keeps a population stable without immigration

Physiologic vs. arithmetic

the bigger the gap, the less of the land is farmable

Density

- **Population density:** people vs. land, as if spread evenly.
- **Arithmetic** = people ÷ *all* land. U.S.: 326 million, **86 per sq mi** (33 per sq km).
- Bangladesh **2,962** · Netherlands **1,068** · Japan **869** per sq mi.
- Misleads: hides empty Alaska and the West, and Egypt's desert.
- **Physiologic** = people ÷ *arable* land.
- **Egypt:** 97.6 million; 252 → **6,995** per sq mi; **98%** of people on **3%** of the land (Nile valley and delta).
- **Switzerland:** physiologic = 10 × arithmetic (1 acre in 10 arable).
- **Ukraine:** 44.8 million; 192 → 281 (× 1.46: 1 of every 1.46 acres arable).
- **Agricultural (AP)** = farmers ÷ farmland: rural Bangladesh very high, Iowa very low.
- Middle America and the Caribbean high; South America moderate; India lowest in South Asia and has more farmland per person than China.

Population data

- U.S. census every **10 years**: money and seats in Congress depend on it.
- Undercounted: migrants, racial minorities, doubled-up or homeless families, low-income families.
- Debates: **sampling** vs. counting everyone; the **citizenship question**.
- UN, World Bank, Population Reference Bureau: numbers don't always match; often **estimates**.

Distribution and clusters

- **Distribution:** the pattern people make across space. **Clustered** = close together; **dispersed** = few, spread out.
 - People lived near food; farm technology and shipping loosened that link.
 - **Dot map** (Fig. 2.4: 1 dot = 100,000 people). **Dasymetric map** (Fig. 2.5) uses land cover and terrain.
 - China + India = $\frac{1}{2}$ of humanity (empty Himalayas, China's desert).
1. **East Asia:** almost $\frac{1}{4}$ of humanity, 1.4 billion in China; Huang He and Yangtze ribbons; wheat and rice.
 2. **South Asia:** 1.5 billion; India, Pakistan, Bangladesh, Sri Lanka; Ganges, Indus, Brahmaputra; hemmed in by Himalayas, Afghan/Pakistani mountains, Indian Ocean. India 1.17 vs. China 0.56; India most populous by 2030 (1 in 6).
 3. **Europe:** 725+ million; dense even in mountains; Industrial Revolution first; urban: Germany 77%, UK 83%, France 80%.
 4. **North America:** megalopolis Boston–Washington, D.C. and Quebec City–Windsor; 70+ million; half of Canada; 20% of Americans; $\frac{1}{4}$ the size of Europe's cluster.
- Big three ≈ **4 billion** of 7.7 billion. South America + Africa barely beat India.
 - Dhaka **122,700** · Mumbai **68,400** · NYC **28,717** per sq mi. Rural Bangladesh 3,000–5,000 vs. rural Iowa **55**.

Opening and big ideas

- **Shanghai:** 128-story tower (2,000+ ft); ~22 million; China 1.4 billion, 100+ cities over 1 million.
- Coal → **PM-2.5** (reaches lungs and blood). North China lives **5.5 years** less. Fixes: limit coal, ban imported plastics for burning, limit cars.
- **Demography** = study of population trends.
- **Scale:** downtown → suburbs vanishes on a world map; national averages hide states.
- **Yanゴン:** density + poverty + weak infrastructure = vulnerability; 2008 cyclone, Irrawaddy delta, ~100,000 dead.
- **Luxor:** environment shapes where people live, not just culture.

Malthus and his critics

- **Ehrlich**, *The Population Bomb* (late 1960s): population outpacing food.
- **Malthus**, 1798: food grows **linearly**, population **exponentially**; revised 1803–1826; assumed a country eats what it grows.
- **Wrong because:** countries aren't closed systems (globalization; mercantilism, colonialism, capitalism); crops spread (potato → Ireland, 1700s); Norway 2% arable, imports food; yields rose (seeds, pesticides, fertilizers, irrigation, GMOs).
- **Neo-Malthusians:** focus on population; limits of food, energy and water; farming's environmental costs.

Growth around the world

- **CBR/CDR** = births/deaths per 1,000 per year. NIR leaves out migration. **Negative:** Europe and Russia.
- Soviet growth fell after **1991**. India nearly 3% thirty years before the book; now Africa **2.43%** vs. India **1.17%**; HIV/AIDS.
- Palestinian territories 2.83 · Afghanistan 2.65 · Sudan 2.55 · Yemen 2.52. Iran, Oman, Morocco dropping.
- Limited women's opportunities + men running family life → high growth.
- India below world avg. but above China; Japan shrinking; Thailand negative; Indonesia and Vietnam slowing.
- South America a bit over 1%; Brazil **2.9%** → **0.8%**; Argentina, Chile, Uruguay low.
- Lowest: **global economic core** (U.S., Canada, Europe, Japan, Australia, Uruguay).
- Russia after 1991: health, alcoholism, drugs, male suicide, economy. Ukraine negative.
- More urban → lower natural increase.

World population

2,000 years ago	250 million
1650	500 million
1820	1 billion
1900	1.6 billion
1930	2 billion
1975	4 billion
2000	6.1 billion
2019	7.7 billion

- Doubling time only **39 years** (mid-1980s) → **54** now. \$100 at 10% doubles in ~7 years.
- Life expectancy **30** (1900) → **72** (2016).
- Levels off at **10–11 billion**, 2050–2100. Now ~1.2% a year, 80+ million a year.

Fertility and aging

- **TFR**: average children per woman 15–49. **95+ countries** (41% of people) below 2.1.
- 2016: world **2.4** (the book also says 2.43), Niger **7.2**, South Korea **1.2**. UN: below 2.2 by 2050.
- Drops: Kenya 3.85; China 6.1 → 1.75 in 35 yrs, 1.5 (2010), 1.62 (2016); Iran 6.8 (1980) → 1.7 (2016).
- **Old-age dependency**: Europe 29.9 → 47 by 2050; sub-Saharan Africa 5.7. Child dependency: Africa 74 vs. Europe 23.
- Aging = pensions and care, fewer young taxpayers. Fixes: **raise TFR** or **immigration**.
- **Japan**: 128.06M (2008) → 126.8M (2017) → ~100M (2050, ~20%); discourages immigration; 98%+ Japanese.
- Raising births: Sweden, Russia (leave, daycare); Japan (men's housework). Limited success.
- Still growing because of India, Indonesia, Bangladesh, Pakistan, Nigeria.

India

- 29 states + 7 union territories. North grows fast; south and west slow (female literacy, land ownership, health care, birth control).
- **1952**: first population program in the world.
- **1976**: forced sterilization of men with 3+ children; quotas; Maharashtra **3.7 million**; riots; **6 million** men in all.
- Now: states set policy (Maharashtra pays newlyweds to wait 2 years); birth control for women, injectables, posters, village clinics.
- **Contraceptive prevalence** (women 15–49): India **55.1%**; Andhra Pradesh 69.8, Maharashtra 63.5, Bihar 26.0, Meghalaya 21.1.
- Lesson: a world map hides what's inside a country.

Religion and growth

- Catholic Church and some conservative Islam oppose contraception. **Northern Nigeria** grows faster than the south.
- But the Catholic heartland grows slowly; the **Philippines** grows fast.
- **Bordeaux** (de Blij, from Dakar): "Where are all the children?" Europe is aging.
- Saudi Arabia ~2%; Indonesia's program from **1970** (coercion + inducement) → 1.6% (2000), 1.1% (2016).

Pyramids



	Expansive	Stationary (chimney)	Constrictive
Births	High TFR	Low, ~2.1 or less	Very low
Deaths	High infant deaths, short lives	Long lives	Long lives; deaths rising
Growth	Fast	Slow / stable	Shrinking
Income	Lower	Higher	Wealthy, aging
Burden	Children (Africa 74)	Rising old-age	Few young taxpayers
DTM	2	4	5
e.g.	Niger, Guatemala	France, Italy, Sweden	Japan

- Males left, females right; 5-year groups, youngest at bottom; **check the top bracket**.
- **Lower-income**: ages 0–14 over **40%**, top three groups under **10%**. Pakistan, Yemen, Guatemala, Cameroon, Laos.
- **Composition** = makeup by age, sex, marital status, education.

Demographic transition model

Stage	Births	Deaths	Growth	Examples and causes
1	High	High, swinging	Low	Most of history. Black Death , 1300s: Crimea → Sicily → north on rats' fleas (contagious diffusion); killed ¼–½; Britain ~4M → ~2M. Famine and war.
2	High	Falling fast	Very high	Europe after ~1750: food, sanitation, soap, medicine. Death rate ~35 (Europe, before 1750) → ~16 (Britain, 1850). Today: Afghanistan, Palestine (vaccines).
3	Falling fast	Falling	Low, still growing	Britain ~1870 through the world wars: women's opportunities, later marriage, fewer child deaths. Brazil (contraceptives); Mali (schooled girls have ~half as many children).
4	Low	Low	Stable or slowing	Britain after 1950: contraception, legal abortion. U.S., U.K. today.
5	Very low	Rising	Falling, negative	Japan, Russia . Zero population growth in ~50 years → a stationary population level (SPL).

Based on western Europe after the Industrial Revolution. Late-1980s World Bank SPL predictions (U.S. 276M by 2035, Brazil 353M by 2070, Mexico 254M by 2075, China 1.4B by 2090, India 1.6B by 2150) were unrealistic: China passed 1.2B in 1994, India 1B in 1998. Newer: China **1.45B in 2030**, India **1.7B in 2060**.

Common traps

- Rates are **per 1,000**: divide by 10 for a percent (25 per 1,000 = 2.5%, not 25%).
- **Stage 2 vs. 3**: if births have started falling, it's stage 3.
- **Natural increase leaves out migration**: a negative NIR country can still grow through immigration.
- **Arithmetic vs. physiologic**: pressure on farmland is physiologic.
- **¼ vs. ½**: East Asia is ~¼ of humanity; China + India together are ½.
- **55 vs. 86**: rural Iowa is 55 per sq mi; the U.S. average is 86.
- **3% vs. 2%**: Egypt's people live on 3% of the land; Norway's land is 2% arable.
- **1952 vs. 1976**: India's first program vs. the sterilization drive.
- **2.4 vs. 2.43**: the book gives both for the 2016 world TFR.
- **Over 40%**: the text says ages 0–14 are over 40% in most lower-income countries; use that, not the chart.
- **Religion isn't destiny**: the Catholic heartland grows slowly.